2.01 Trigonometry: Radians, Arcs and Sectors

Worked Solutions 1–25

1)

Degrees	Radians
$180°$	π
$90°$	$\frac{\pi}{2}$
$60°$	$\frac{\pi}{3}$
$45°$	$\frac{\pi}{4}$
$36°$	$\frac{\pi}{5}$
$30°$	$\frac{\pi}{6}$
$20°$	$\frac{\pi}{9}$
$160°$	$\frac{8\pi}{9}$
$1.62°$	$\frac{1.62\pi}{180} = \frac{9\pi}{1000}$
$1.62 \times \frac{180}{\pi} \approx 92.8°$	1.62

2 a)

Arc Length $= \theta r = 0.5 \times 10 = 5$ cm

2 b)

Arc Length $= \theta r = \frac{\pi}{4} \times 20 = 5\pi$ m

2 c)

Arc Length $= \theta r = 0.75 \times 15 = 11.25$ cm

2 d)

Arc Length $= \theta r = \frac{3\pi}{8} \times 120 = 45\pi$ m

3 a)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times 0.4 \times 20^2 = 80$ cm^2

3 b)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times \frac{\pi}{3} \times 9^2 = \frac{27\pi}{2}$ m^2

3 c)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times 1.15 \times 64^2 = 2355.2$ cm^2

3 d)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times \frac{\pi}{20} \times 18.5^2 = \frac{1369\pi}{160}$ m^2

4 a)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times 0.8 \times 8^2 = 25.6$ cm^2

Arc Length $= \theta r = 0.8 \times 8 = 6.4$ cm

Perimeter $= 2r + \theta r = 2 \times 8 + 6.4 = 22.4$ cm

4 b)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times \frac{\pi}{5} \times 14^2 = \frac{98\pi}{5}$ m^2

Arc Length $= \theta r = \frac{\pi}{5} \times 14 = \frac{14\pi}{5}$ m

Perimeter $= 2r + \theta r = 2 \times 14 + \frac{14\pi}{5} = (28 + \frac{14\pi}{5})$ m $= 36.8$ m to 3 sf

4 c)

Area $= \frac{1}{2}\theta r^2 = \frac{1}{2} \times 2.95 \times 6.5^2 = \frac{9971}{160}$ cm^2 $= 62.31875$ cm^2

Arc Length $= \theta r = 2.95 \times 6.5 = \frac{767}{40}$ cm $= 19.175$ cm

Perimeter $= 2r + \theta r = 2 \times 6.5 + 19.175 = \frac{1287}{40}$ cm $= 32.175$ cm

4 d)

Area $= \frac{1}{2} \times \frac{11\pi}{18} \times 360^2 = 39600\pi$ m^2

Arc Length $= \theta r = \frac{11\pi}{18} \times 360 = 220\pi$ m

Perimeter $= 2r + \theta r = 2 \times 360 + 220\pi = (720 + 220\pi)$ m $= 1410$ m to 3 sf

5)

Perimeter $= 25 \Rightarrow 2 \times 8 + \theta \times 8 = 25 \Rightarrow 8\theta = 9 \Rightarrow \theta = \frac{9}{8}$

6)

Area $= 50 \Rightarrow \frac{1}{2}\theta r^2 = 50 \Rightarrow r^2 = \frac{100}{\theta} \Rightarrow r = \sqrt{\frac{100}{\theta}}$ $(= \frac{10}{\sqrt{\theta}})$

7)

Perimeter $= 84 \Rightarrow 2r + \theta r = 84 \Rightarrow \theta r = 84 - 2r \Rightarrow \theta = \frac{84-2r}{r}$ $(= \frac{84}{r} - 2)$

8)

Area of $OCD = \frac{1}{2} \times \frac{\pi}{6} \times 9^2 = \frac{27\pi}{4}$

$$\frac{27\pi}{4} = 3 \times (\frac{1}{2} \times \frac{\pi}{6} \times OA^2) \Rightarrow \frac{9\pi}{4} = \frac{\pi}{12} \times OA^2 \Rightarrow OA^2 = 27 \Rightarrow OA = 3\sqrt{3} \text{ cm}$$

9 a)

Area of Sector $= \frac{1}{2} \times 0.9 \times 6^2 = \frac{81}{5}$ cm^2 $= 16.2$ cm^2

Area of Triangle $= \frac{1}{2} \times 6 \times 6 \times \sin 0.9 = 14.0998\ldots$ cm^2

Area of Shaded Segment $= 16.2 - 14.0998\ldots = 2.1001\ldots$ cm^2 $= 2.10$ cm^2 to 3 sf

9 b)

Arc Length $= 0.9 \times 6 = 5.4$ cm

$$x^2 = 6^2 + 6^2 - 2 \times 6 \times 6 \times \cos 0.9 \Rightarrow x^2 = 27.2440\ldots \Rightarrow x = 5.2195\ldots \text{ cm}$$

Perimeter of Shaded Segment $= 5.4 + 5.2195\ldots = 10.6195\ldots$ cm $= 10.6$ cm to 3 sf

10 a)

$$x^2 + y^2 - 4x + 14y + 19 = 0 \Rightarrow (x-2)^2 - 4 + (y+7)^2 - 49 + 19 = 0 \Rightarrow (x-2)^2 + (y+7)^2 = 34$$

Radius $= \sqrt{34}$

Area of Sector $= \frac{1}{2} \times 2.25 \times (\sqrt{34})^2 = \frac{153}{4} = 38.25$

Area of Triangle $= \frac{1}{2} \times \sqrt{34} \times \sqrt{34} \times \sin 2.25 = 13.2272\ldots$

Area of Shaded Segment $= 38.25 - 13.2272\ldots = 25.0227\ldots$ units^2 $= 25.0$ units^2 to 3 sf

10 b)

Area of Circle $= \pi \times (\sqrt{34})^2 = 34\pi$

Percentage Shaded $= \frac{25.0227\ldots}{34\pi} \times 100 = 23.4264\ldots\% = 23.4\%$ to 3 sf

11)

Area of Sector $= \frac{1}{2} \times \frac{2\pi}{7} \times 10^2 = \frac{100\pi}{7}$ m^2

As C is the midpoint of O and B, $OC = 5$ m.

Area of Triangle $= \frac{1}{2} \times 10 \times 5 \times \sin \frac{2\pi}{7} = 19.5457\ldots$ m^2

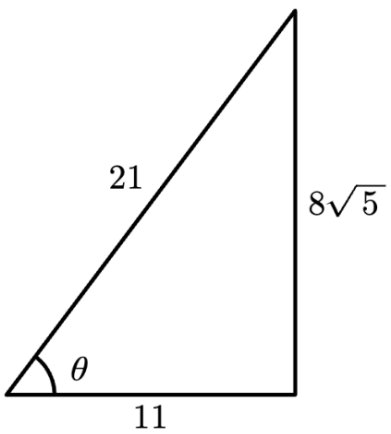
Area of Shaded Segment $= \frac{100\pi}{7} - 19.5457\ldots = 25.3341\ldots$ m^2 $= 25.3$ m^2 to 3 sf

12 a)

$$8^2 = 9^2 + 7^2 - 2 \times 9 \times 7 \times \cos\theta \Rightarrow 64 = 130 - 126\cos\theta \Rightarrow \cos\theta = \frac{130-64}{126} = \frac{66}{126} = \frac{11}{21}$$

12 b)

$\cos\theta = \frac{11}{21}$. θ is **ACUTE**. Draw a triangle with this information:



$\sin\theta = \frac{8\sqrt{5}}{21}$.

Alternatively:

$$\sin^2\theta + \cos^2\theta = 1 \Rightarrow \sin^2\theta = (\tfrac{11}{21})^2 = 1 \Rightarrow \sin^2\theta = \tfrac{320}{441} \Rightarrow \sin\theta = \tfrac{8\sqrt{5}}{21}$$

12 c)

$\theta = \arccos\frac{11}{21} = 1.0194\ldots$

Area of Sector $= \frac{1}{2} \times (1.0194\ldots) \times 9^2 = 41.2889\ldots$ cm^2

Area of Triangle $= \frac{1}{2} \times 9 \times 7 \times \frac{8\sqrt{5}}{21} = 12\sqrt{5}$ cm^2

Area of Shaded Segment $= 41.2889\ldots - 12\sqrt{5} = 14.4560\ldots$ cm^2 $= 14.5$ cm^2 to 3 sf

13)

Area of Triangle $= \frac{1}{2} \times 15 \times 5\sqrt{5} \times \sin 1.1 = 74.7300\ldots$ cm^2

Area of Sector $= 74.7300\ldots - 20 = 54.7300\ldots$ cm^2

$$54.7300\ldots = \tfrac{1}{2} \times 1.1 \times r^2 \Rightarrow r^2 = 99.5091\ldots \Rightarrow r = 9.9754\ldots \text{ cm}$$

Arc Length $= 1.1 \times 9.9754\ldots = 10.9729\ldots$ cm

$$AB^2 = 15^2 + (5\sqrt{5})^2 - 2 \times 15 \times (5\sqrt{5})\cos 1.1 \Rightarrow AB^2 = 197.8592\ldots \Rightarrow AB = 14.0662\ldots \text{ cm}$$

Perimeter of Shaded Region $= 14.0662\ldots + (15 - 9.9754\ldots) + (5\sqrt{5} - 9.9754\ldots) + 10.9729\ldots =$ $31.2687\ldots$ cm $= 31.3$ cm to 3 sf

14 a)

Area of Sector $= 2 \Rightarrow \frac{1}{2}\theta r^2 = 2$. Perimeter $= 6 \Rightarrow 2r + \theta r = 6 \Rightarrow \theta r = 6 - 2r \Rightarrow \theta = \frac{6}{r} - 2$.

$$\tfrac{1}{2}(\tfrac{6}{r} - 2)r^2 = 2 \Rightarrow (\tfrac{6}{r} - 2)r^2 = 4 \Rightarrow 6r - 2r^2 = 4 \Rightarrow 2r^2 - 6r + 4 = 0 \Rightarrow r^2 - 3r + 2 = 0$$

14 b)

$(r-2)(r-1) = 0 \Rightarrow r = 2$ or $r = 1$. If $r = 2$, then $\theta = \frac{6}{2} - 2 = 1$. If $r = 1$, then $\theta = \frac{6}{1} - 2 = 4$.

15)

Area of Sector $= 45 \Rightarrow \frac{1}{2}\theta r^2 = 45$. Perimeter of Sector $= 2r + \theta r$.

$$2r + \theta r = 6 \times \theta r \Rightarrow 5\theta r = 2r \Rightarrow 5\theta = 2 \Rightarrow \theta = \tfrac{2}{5}$$

If $\theta = \frac{2}{5}$, then $\frac{1}{2} \times \frac{2}{5} \times r^2 = 45 \Rightarrow r^2 = 225 \Rightarrow r = 15$.

16)

Area of Triangle $= \frac{1}{2} \times 8 \times 8 \times \sin 1.4 = 31.5343\ldots$ cm^2

$$AB^2 = 8^2 + 8^2 - 2 \times 8 \times 8 \times \cos 1.4 \Rightarrow AB^2 = 106.2442\ldots \Rightarrow AB = 10.3074\ldots \text{ cm}$$

Radius of Semi-Circle $= \frac{1}{2} \times 10.3074\ldots = 5.1537\ldots$ cm

Area of Semi-Circle $= \frac{1}{2}\pi \times (5.1537\ldots)^2 = 41.7220\ldots$ cm^2

Area of first toy $= 31.5343\ldots + 41.7220\ldots = 73.2563\ldots$ cm^2

$$\tfrac{1}{2} \times 1.4 \times r^2 = 73.2563\ldots \Rightarrow r^2 = 104.6519\ldots \Rightarrow r = 10.2299\ldots = 10.2 \text{ cm to 3 sf}$$

17)

Area of logo $= 2 \Rightarrow \frac{1}{2}\theta r^2 + \frac{1}{2}\theta(\frac{1}{2}r)^2 = 2 \Rightarrow \frac{5}{8}\theta r^2 = 2$. Perimeter $= 9.2$.

$$r + \theta r + \tfrac{1}{2}r + \theta(\tfrac{1}{2}r) + \tfrac{1}{2}r = 9.2 \Rightarrow 2r + \tfrac{3}{2}\theta r = 9.2 \Rightarrow \tfrac{3}{2}\theta r = 9.2 - 2r$$

$$\theta = \frac{9.2-2r}{\frac{3}{2}r} \Rightarrow \theta = \frac{92}{15r} - \frac{4}{3}$$

$$\frac{5}{8}(\frac{92}{15r} - \frac{4}{3})r^2 = 2 \Rightarrow (\frac{92}{15r} - \frac{4}{3})r^2 = \frac{16}{5}$$

$$\frac{92r}{15} - \frac{4}{3}r^2 = \frac{16}{5} \Rightarrow 92r - 20r^2 = 48 \Rightarrow 20r^2 - 92r + 48 = 0 \Rightarrow 5r^2 - 23r + 12 = 0$$

$$(5r-3)(r-4) = 0 \Rightarrow r = \tfrac{3}{5} \text{ or } r = 4$$

If $r = \frac{3}{5}$, then $\theta = \frac{92}{15\times\frac{3}{5}} - \frac{4}{3} = \frac{80}{9}$. If $r = 4$, then $\theta = \frac{92}{15\times 4} - \frac{4}{3} = \frac{1}{5}$. $\theta = \frac{80}{9} = 8.8888\ldots > 2\pi$ so this value of θ does not give a sector of a circle. So the only valid pair of values are $r = 4$ and $\theta = \frac{1}{5}$.

18)

As B is the midpoint of AC, and triangle OAC is equilateral, we must have that $\angle AOB = \angle BOC = \frac{\pi}{6}$.

Area of the whole triangle $= \frac{1}{2} \times r \times r \sin\frac{\pi}{3} = \frac{\sqrt{3}}{4}r^2$.

$$\cos\tfrac{\pi}{6} = \frac{OB}{r} \Rightarrow OB = \frac{\sqrt{3}}{2}r$$

Area of Sector $OBF = \frac{1}{2} \times \frac{\pi}{6} \times (\frac{\sqrt{3}}{2}r)^2 = \frac{\pi}{16}r^2$.

$$OE = \tfrac{4}{5}OB = \tfrac{4}{5} \times \tfrac{\sqrt{3}}{2}r = \tfrac{2\sqrt{3}}{5}r$$

Area of Sector $ODE = \frac{1}{2} \times \frac{\pi}{6} \times (\frac{2\sqrt{3}}{5}r)^2 = \frac{\pi}{25}r^2$.

Shaded Area $= \frac{\sqrt{3}}{4}r^2 - \frac{\pi}{16}r^2 - \frac{\pi}{25}r^2 = (\frac{1}{4}\sqrt{3} - \frac{41}{400}\pi)r^2$.

19)

Area of Triangle $= \frac{1}{2} \times 5 \times 5 \times \sin\frac{\pi}{3} = \frac{25\sqrt{3}}{4}$.

Area of Sector $= \frac{1}{2} \times \frac{\pi}{3} \times 5^2 = \frac{25\pi}{6}$.

Area of a Segment $= \frac{25\pi}{6} - \frac{25\sqrt{3}}{4}$.

Total Area $= \frac{25\sqrt{3}}{4} + 3(\frac{25\pi}{6} - \frac{25\sqrt{3}}{4}) = (\frac{25\pi}{2} - \frac{25\sqrt{3}}{2})$ cm^2 $= \frac{25}{2}(\pi - \sqrt{3})$ cm^2.

20)

$12\pi = \frac{2\pi}{5} \times OP \Rightarrow OP = 30$. Area of Sector $OPQ = \frac{1}{2} \times \frac{2\pi}{5} \times 30^2 = 180\pi$. Angle $AOC = \frac{\pi - \frac{2\pi}{5}}{2} = \frac{3\pi}{10}$. Area of Sector $OAC = 90\pi$.

$$\tfrac{1}{2} \times \tfrac{3\pi}{10} \times OA^2 = 90\pi \Rightarrow OA = 10\sqrt{6}$$

Therefore $AB = 20\sqrt{6}$ m.

21)

$$x^2 + y^2 - 6x + 12y - 4 = 0 \Rightarrow (x-3)^2 - 9 + (y+6)^2 - 36 - 4 = 0 \Rightarrow (x-3)^2 + (y+6)^2 = 49$$

Therefore $AO = 7 \Rightarrow AB = 14$. As AB is a diameter and C lies on the circle, $\angle ACB = \frac{\pi}{2}$.

$$\cos 0.6 = \frac{AC}{AB} \Rightarrow \cos 0.6 = \frac{AC}{14} \Rightarrow AC = 14\cos 0.6 = 11.5546\ldots$$

Due to symmetry, $AD = AC = 11.5546\ldots$. Area of Sector $ADE = \frac{1}{2} \times 0.6 \times (11.5546\ldots)^2 = 40.0533\ldots$ units^2 $= 40.1$ units^2 to 3 sf.

22)

$$\tan 0.24 = \frac{2}{OA} \Rightarrow OA = \frac{2}{\tan 0.24} = 8.1727\ldots \text{ m}$$

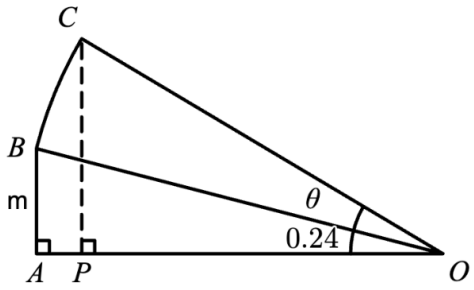
Area of Triangle $OAB = \frac{1}{2} \times 8.1727\ldots \times 2 = 8.1727\ldots$ m^2.

$$OB^2 = (8.1727\ldots)^2 + 2^2 \Rightarrow OB^2 = 70.7932\ldots \Rightarrow OB = 8.4138\ldots \text{ m}$$

Area of Sector $OBC = 8.1727\ldots$.

$$8.1727\ldots = \frac{1}{2} \times \theta \times (8.4138\ldots)^2 \Rightarrow \theta = 0.2308\ldots$$

Drop a perpendicular from C to the line OA. Where this intersects OA, we shall call P.



$$\sin(0.24 + 0.2308\ldots) = \frac{CP}{OC} \Rightarrow \sin(0.4708\ldots) = \frac{CP}{8.4138\ldots}$$
$$CP = 8.4138\ldots \times \sin(0.4708\ldots) \Rightarrow CP = 3.8172\ldots = 3.82 \text{ m to 3 sf}$$

23 a)

Distance between $(5, 5)$ and $(17, 14)$ is $\sqrt{(17-5)^2 + (14-5)^2} = \sqrt{225} = 15$. Radii of the two circles are 5 and 10. $5 + 10 = 15$ so the two circles intersect at a single point.

23 b)

Let θ_1 be the sector angle for $(x-5)^2 + (y-5)^2 = 25$.

$$\tan\theta_1 = \frac{17-5}{14-5} = \frac{12}{9} = \frac{4}{3} \Rightarrow \theta_1 = \arctan\frac{4}{3} = 0.9272\ldots$$

Let θ_2 be the sector angle for $(x-17)^2 + (y-14)^2 = 100$.

$$\theta_2 = \pi - \frac{\pi}{2} - \theta_1 = \frac{\pi}{2} - 0.9272\ldots = 0.6435\ldots$$

Area of Sector1 $= \frac{1}{2} \times 0.9272\ldots \times 5^2 = 11.5911\ldots$. Area of Sector2 $= \frac{1}{2} \times 0.6435\ldots \times 10^2 = 32.1750\ldots$. Area of Triangle $= \frac{1}{2} \times (17-5) \times (14-5) = 54$. Shaded Area $= 54 - 11.5911\ldots - 32.1750\ldots = 10.2337\ldots$ units^2 $= 10.2$ units^2 to 3 sf.

24)

Without loss of generality, we can assume that the right-hand side of the square that passes through the origin O is parallel to the y-axis. Then the top side of the square has equation $y = 4$, $-8 \le x \le 0$. Find where $y = 4$ intersects the circle:

$$x^2 + 4^2 = 36 \Rightarrow x^2 = 20 \Rightarrow x = -2\sqrt{5}$$

So $(-2\sqrt{5}, 4)$ and $(-2\sqrt{5}, -4)$ are the points where the square intersects the circle. The angle from $(0, 0)$ to $(-2\sqrt{5}, 4)$ is $\arctan(\frac{4}{2\sqrt{5}}) = 0.7297\ldots$ to the horizontal.

Remaining circumference of the circle $= (2\pi - 2 \times 0.7297\ldots) \times 6 = 28.9423\ldots$.

Perimeter $= 28.9423\ldots + 8 + 2 \times (8 - 2\sqrt{5}) = 43.9981\ldots$ units $= 44.0$ units to 3 sf.

25)

The line of symmetry of the two overlapping circles (apart from $y = 0$) is $x = \frac{1}{2}k$. This line intersects the circle $x^2 + y^2 = 16$ when

$$(\tfrac{1}{2}k)^2 + y^2 = 16 \Rightarrow y^2 = 16 - \tfrac{1}{4}k^2 \Rightarrow y = \pm\sqrt{16 - \tfrac{1}{4}k^2}$$

As the total perimeter is 10π, the remaining perimeter of one of the circles is 5π. The circumference of each circle is $2\pi \times 4 = 8\pi$, so each circle is missing 3π of circumference (the dashed part of the circle).

$3\pi = \theta \times 4 \Rightarrow \theta = \frac{3\pi}{4}$ is the angle of the sector

Considering the triangle:

$$(2\sqrt{16 - \tfrac{1}{4}k^2})^2 = 4^2 + 4^2 - 2 \times 4 \times 4 \times \cos\tfrac{3\pi}{4}$$

$$4(16 - \tfrac{1}{4}k^2) = 32 + 16\sqrt{2} \Rightarrow 16 - \tfrac{1}{4}k^2 = 8 + 4\sqrt{2} \Rightarrow k^2 = 32 - 16\sqrt{2}$$

Therefore $k = \sqrt{32 - 16\sqrt{2}} = 4\sqrt{2 - \sqrt{2}}$.